Vehicle Immobilization System — Portfolio Report

FPGA / Verilog / Nexys 4 DDR / Xilinx Vivado

1. Overview

The Vehicle Immobilization System is a digital security prototype implemented on a Digilent Nexys 4 DDR FPGA. The project explores PIN-based authorization, digital control logic, and visible access-status indication.

2. Hardware Interface

Interface	Implementation
FPGA board	Digilent Nexys 4 DDR
HDL	Verilog
PIN input	SW0–SW3
PIN	1010
Reset	BTNC, active-low
Output	LED0–LED15

3. Final RTL Behavior

Correct PIN (1010) produces 16'hFFFF, giving an all-LED authorized indication. Any other four-bit input produces 16'hAAAA, giving an alternate LED unauthorized indication. Active-low BTNC reset clears the LED output.

4. Version Reconciliation

The original academic report describes a confirmation-button/FSM/attempt-count concept. The supplied top-level RTL connects btnC to reset, and the supplied XDC exposes btnC as the reset input with no separate confirmation input. The final public RTL therefore follows the supplied hardware interface instead of silently mixing incompatible versions. The discrepancy is documented in docs/reconciliation.md and the original academic report is retained unchanged.

5. Academic Evidence

The submitted report identifies the project as a Digital System Design mini project at Manipal Institute of Technology during Aug–Dec 2024. It documents the Nexys 4 DDR platform, PIN 1010, LED behavior, methodology, design/truth table, and physical implementation photographs.

6. Team Contribution

The original report records Darshit Chheda's contribution as testing and debugging, alongside Verilog development by Pisupati Rama Sridhrut and documentation/system-design work by Satwik Abhay.

7. Limitations

This is an FPGA laboratory prototype and not an automotive-grade immobilization system. The repository should not be used as instructions for connecting the design to a real vehicle.